

# Algebraic perfect hash functions

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# Chapter 1

## Algebraic perfect hash functions

See the function reference

A *set* is an unordered collection of distinct elements. If we know the elements in a set are given by  $a$ ,  $b$ , and  $c$ , we may denote the set with the notation  $\{a, b, c\}$ .

A *finite* set has a finite number of elements. For example,  $\{1, 3, 5\}$  is a finite set with three elements. When sets  $\mathbb{A}$  and  $\mathbb{B}$  are *isomorphic*, denoted by  $\mathbb{A} \cong \mathbb{B}$ , they can be put into a one-to-one correspondence (bijection), e.g.,  $\{b, a, c\} \cong \{1, 2, 3\}$ . Since there exists at least one bijection between isomorphic sets, we can losslessly convert one to the other and thus, isomorphic sets are in some sense equivalent.

The cardinality of a finite set  $\mathbb{A}$  is the number of elements in the set, denoted by  $|\mathbb{A}|$ , e.g.,  $|\{1, 3, 5\}| = 3$ . A *countably infinite* set is isomorphic to the set of natural numbers  $\mathbb{N} = \{1, 2, 3, \dots\}$ .

Given two elements  $a$  and  $b$ , an ordered pair of  $a$  then  $b$  is denoted by  $(a, b)$ , where  $(a, b) = (c, d)$  iff  $a = c$  and  $b = d$ . Ordered pairs are non-commutative and non-associative, i.e.,  $(a, b) \neq (b, a)$  if  $a \neq b$  and  $(a, (b, c)) \neq ((b, a), c)$ .

Related to the ordered pair is the Cartesian product.

**Definition 1.1.** The set

$$\mathbb{X} \times \mathbb{Y} := \{(x, y) : x \in \mathbb{X} \wedge y \in \mathbb{Y}\}$$

is the Cartesian product of sets  $\mathbb{X}$  and  $\mathbb{Y}$ .

By the non-commutative and non-associative property of ordered pairs, the Cartesian product is non-commutative and non-associative. However, they are isomorphic, i.e.,  $\mathbb{X} \times \mathbb{Y} \cong \mathbb{Y} \times \mathbb{X}$ .

A *tuple* is a generalization of order pairs which can consist of an arbitrary number of elements, e.g.,  $(x_1, x_2, \dots, x_n)$ .

**Definition 1.2.** The  $n$ -fold Cartesian product of sets  $\mathbb{X}_1, \dots, \mathbb{X}_n$  is given by

$$\mathbb{X}_1 \times \dots \times \mathbb{X}_n := \{(x_1, \dots, x_n) : x_1 \in \mathbb{X}_1 \wedge \dots \wedge x_n \in \mathbb{X}_n\}.$$

Note that

$$\mathbb{X}_1 \times \mathbb{X}_2 \times \mathbb{X}_3 \cong \mathbb{X}_1 \times (\mathbb{X}_2 \times \mathbb{X}_3) \cong (\mathbb{X}_1 \times \mathbb{X}_2) \times \mathbb{X}_3,$$

thus we may implicitly convert between them without ambiguity.

If each set in the  $n$ -ary Cartesian product is the same, the power notation may be used, e.g.,  $\mathbb{X}^3 \equiv \mathbb{X} \times \mathbb{X} \times \mathbb{X}$ . As special cases, the 0-ary (nullary) Cartesian product is defined to be  $\{\emptyset\}$  and the 1-ary (unary) Cartesian product is the identity, e.g.,  $\mathbb{X}^1 = \mathbb{X}$ .

**Definition 1.3.** A one-to-one relation between  $\{0, 1\}^m$  and  $\{1, 2, \dots, 2^m\}$  is given by the equation

$$b_m \iff \sum_{j=1}^m 2^{m-j+1} b_j,$$

where  $b_m \in \{0, 1\}^m$ . We denote the transformation in either direction by the unary operation  $'$ , e.g.,  $2' = 0$ .

We show some of the mappings in Table 1.1.

Table 1.1: A one-to-one relation between  $\{0, 1\}^*$  and  $\mathbb{N}$ .

| $\mathbb{N}$ | $\{0, 1\}^*$ |
|--------------|--------------|
| 1            | $\emptyset$  |
| 2            | $0_2$        |
| 3            | $1_2$        |
| 4            | $00_2$       |
| 5            | $01_2$       |
| 6            | $10_2$       |
| 7            | $11_2$       |
| 8            | $000_2$      |
| 9            | $001_2$      |
| 10           | $010_2$      |
| 11           | $011_2$      |
| 12           | $100_2$      |
| 13           | $101_2$      |
| 14           | $110_2$      |
| 15           | $111_2$      |

$$\frac{\mathbb{N} \quad \{0, 1\}^*}{\vdots \quad \vdots}$$

A *hash function* is given by the following definition.

**Definition 1.4.** Hash functions of type  $\mathbb{X} \mapsto \mathbb{Y}$  are just total functions, normally with a finite codomain, with the weak assumption that they will be used as a device to assign elements from  $\mathbb{X}$  a value from  $\mathbb{Y}$  without requiring any particular rule.

For a given bit string  $x$  and hash function  $\text{hash}$ ,  $y = \text{hash}(x)$  is denoted the *hash* of  $x$ .

We are particularly interested in *perfect* hash functions as given by the following definition.

::: {definition name=“Perfect hash function”} A perfect hash function over the set  $\mathbb{A} \subseteq \mathbb{X}$ , denoted by

$$\text{hash}_{\mathbb{A}} : \mathbb{X} \mapsto \mathbb{Y},$$

is an injective function when restricted to  $\mathbb{A}$ . That is, there are no collisions among elements of  $\mathbb{A}$ ,  $\text{hash}_{\mathbb{A}}(x) \neq \text{hash}_{\mathbb{A}}(y)$  for all  $x, y \in \mathbb{A}$ ,  $x \neq y$ . \end{definition} Perfect hash functions are generally *surjective* when not restricted to the range of elements it perfectly hashes.

The *load factor* is given by the following definition.

**Definition 1.5** (Load factor). The proportion of hashes mapped to by  $\text{hash}_{\mathbb{A}} : \mathbb{X} \mapsto \mathbb{N}$  over subset  $\mathbb{A}$  is denoted the load factor. Specifically, the load factor of  $\text{hash}_{\mathbb{A}}$  is a rational number given by

$$\frac{|\text{hash}_{\mathbb{A}}(\mathbb{A})|}{|\text{image}(\text{hash}_{\mathbb{A}})|}.$$

A *minimal* perfect hash has a load factor 1.

A perfect hash function of type  $\mathbb{X} \mapsto \mathbb{Y}$  over  $\mathbb{A}$  with a load factor  $r$  may be denoted by  $\text{hash}_{\mathbb{A}}^r$ . If  $m = |\{\mathbb{A}\}|$  and we are interested in drawing attention to the cardinality of the perfect hash function, we may also denote it by  $\text{hash}_m^r$ .

Every hash function in  $\mathbb{X} \mapsto \mathbb{Y}$  is a perfect hash function over some subset of  $\mathbb{X}$ , e.g., every hash function is trivially a perfect hash function of the empty set and singleton sets.

Assuming  $\mathbb{X}$  and  $\mathbb{Y}$  are finite, the set of hash functions of type  $\mathbb{X} \mapsto \mathbb{Y}$ , which may also be denoted by  $\mathbb{Y}^{\mathbb{X}}$ , has a cardinality

$$|\mathbb{Y}^{\mathbb{X}}|.$$

The set of perfect hash functions over  $\mathbb{A} \subseteq \mathbb{X}$  is a subset of  $\mathbb{X} \mapsto \mathbb{Y}$  with the predicate that no collisions may occur on any pair of elements in  $\mathbb{A}$ .

Figure 1.1: A perfect hash function  $\text{hash}_{\mathbb{X}}^{0.6}: \mathbb{U} \mapsto \{0, 1, 2, 3, 4\}$ .

